

I: Okay. All right, so before my first question. I would like to acknowledge that due to covid there's a disruption in teaching. So, for any of the questions I'm about to ask, please feel free to answer with reference with your normal teaching practice or your teaching practices as adapted for the crisis situation, or even considering both. So, my first question is a background question that I would like to understand a little bit about your experience teaching undergrad students. So, could you briefly describe your experience teaching undergrad students including how does your teaching relate to your current or even your past research?

UCSB09: Okay. So, my background is in [major redacted]. My PhD is in [field redacted]. And after my PhD I went and did three years of work at the [organization redacted]. So, my interests are in development training as a quantitative social science [field redacted] more generally, and the interests are already in [area redacted]. You know, [area redacted] in general. Now, I teach in the [department redacted] so it's a little bit unusual because that isn't quite the mix that is covered in [discipline redacted]. That is to say, there are people who do think about [specific topic redacted], there are people who do think about development. But there's actually no other [professional redacted] in the department and there's no other quantitative social scientists in the department at all. So as a result, when it comes to the undergraduates, you know it's a, you know, it's something that I can do. I can expose them to quantitative stuff for one quarter but never really any more than that. So, I teach this one class every year, it's called [title course redacted]. It's taught to juniors and seniors, I have typically about 130 students in the class at a time. And I make it a point in that class to make them at least responsible consumers of quantitative data. That's the goal, right. To learn, to actually teach them anything about how to do quantitative analysis in any serious way is very difficult, because they're not seeing any of it in any other classes, many of them are taking classes where they're actually, you know, actively discouraged from dealing with quantitative social science. And you know they just don't have any training, many of them are math phobic. They have chosen global studies because they are either afraid of mathematics or somehow, you know, have some issue with it. So, that's kind of what I'm dealing with and so you know if you, the approach I take in class is to start out with, you know, big questions that they would want to know the answers to. And then try to start out by showing them some data showing them some correlations showing them, show them, you know, moving pictures of correlations over time. The business website gap minder.com. That's extremely useful for this; it animates scatter plots over time. And so, you can do a lot with that. Show them, you know, countries' infant mortality rates, GDP, etc. Watch these things moving. And then from there, launch into a critical discussion of, you know, what are accurate statements you can make based on these data, then which statements misinterpreting correlation as causation. Why, how do you fix it. Try to get them to read one or two papers that try to do it. And they can't actually read the whole thing, they can't make any sense of the guts of most of this work but they can, as you guide them through the arguments and say you know this is what this approach is trying to do you're comparing, you know, the rate of decrease in infant mortality in the same country over time during periods of high growth and low growth so how does that influence how you know, what does that relationship tell you is that close enough to causal for you to conclude that economic growth reduces infant mortality things like this, right. So, it's really taking data, showing them what the data are, animating it, trying to make statements, showing the logical pitfalls that you can make when interpreting the data, and then

showing them that there are, in principle, ways to get around all this. But that's as far as I can pick any student in my department.

I: Okay, so I'm just to reiterate please correct me if I misunderstood, you teach [class name] and it's only focused on undergrads or you do also accept graduate students in this class?

UCSB09: We also have graduate students. So we have, we have had a master's program for a long time. And by the way, I should say I've been doing this since [year redacted]. So we've had a master's program since [year redacted], and a PhD program since I think [year redacted]. But it doesn't really make much of a difference at the, in terms of the curriculum they have in our department, our department is indifferent to quantitative research, and some faculty are actually hostile to it. So the students get no training in it. And so every once in a while we get a master student or a PhD student who has had some quantitative training as an undergraduate, maybe they majored in [majors redacted]. And then they are here. And then I sometimes end up having to work very closely with them to help to supervise a project to basically train them from the ground up, but it ends up being a solo act and something I'm increasingly keen not to do, it just takes too much time. So, you know, that's why that is but yes PhD students you know that's the place where you can find one or two that come in with some training.

I: Okay. Okay, thank you. In terms of getting access to data, for, for this particular class you teach, how do your students usually obtain the data they engage with? Let's say you provide the data, do you actually ask them to find data sets connected to the topic? How do you usually provide the data?

UCSB09: Oh, yeah, so what I do, honestly is I get the data myself from whatever source, whether it's the world development indicators or some research project that I'm working on, you know, because I'm always working on some data, so I get the data myself. And then I organize it into an Excel file and I have to write very long, sort of instructional point by point instructions for them on how to then take it from there. If they had to download the data and organize it into Excel, nothing would get done. Because I, you know, bear in mind I also often have TAs who have very little quantitative training. And the other point I'll just say quickly is you know when we do these exercises we have to give them stuff that is you know, the food has to be half chewed before it is given to them. Right. Then they'll finish it. And to finish it, they can only do it if we actually go to computer labs, book computer labs for discussion sections. I have to write detailed scripts for my TAs. And they then run it in sections, and then they also have office hours and computer labs and that's how that's how the students get through it.

I: So, you have mentioned some data sources that you usually explore to find data and that you do some work later on. How do you usually find data sets that are connected to your teaching? What are the main data sources you go for?

UCSB09: So, there's the world development indicators. That's the World Bank right. UNDP has data on a lot of things but it tends not to be particularly well organized. So, for teaching I don't take the time to use UNDP very often. And every once in a while I'll take a data set distilled from

one that I'm working on. It's obviously got to be something that's in the public domain so if I've got survey data on something. Or, if there is, you know, a project I've been working on where I have accumulated data from many other sites. And then processed it into a form that an undergraduate would be able to work with without any prior training then I use that. So, one other site is the [website redacted] and development center. Trying to think of what else I have used. I think those are the main ones, there's only so much you can actually do with untrained students, right. So, it's and, you know, the big questions in [class code redacted], we have data from, from the world development indicators, or there's one other thing I do, which is in a different class and, which is a class on [topic redacted]. I have them compile, I start with data sets produced by the [organization redacted] which are just country sheets that sort of give you a country's balance of payments, the [topics redacted], what the currencies have been doing etc. So, I give them a bunch of that information. Just links to excel sheets that have this stuff put together. And then I have them produce PowerPoint presentations where they develop analyses of these countries, and support it with graphs that they make an Excel from these sheets. And for and for that I had to, you know, there was a parent of one of my former students who gave, you know, made a \$2,000 donation to the department so I've been using that every year to hire lab assistants for a few hours to help the students with this.

I: Oh, nice, that was a generous act from the former student. So, you mentioned you usually use Excel sheets to distribute or to make these datasets. So, do you share like a Google Drive with them, with the class and then they have access to the data set.

UCSB09: I don't use Google.

I: Okay.

UCSB09: I try to avoid doing anything related to students on Google. I have very serious misgivings about the university using Google for everything and particularly for student records. Because I have every reason to suspect that Google is one day going to become an employment agency, and going to be selling information about students. I've gone through their privacy policies; they don't protect our students at all, so they might be using someone's performance in our classes to eventually tell employers whether they should hire this person or not. And so, I won't use Google. So, I just use Excel on Gauchospace, I make Excel files and I put them on Gauchospace and that's it.

I: Okay. You mentioned different data sources, number of agencies that you find data, and you use data for your teaching. Would you say or have you experienced any challenges in obtaining data sets, even from the agencies you listed or I recall you mentioned that most of the datasets you use are in the public domain so I believe that not a recurrent thing you experience very often, but have you ever experienced any issues to find data?

UCSB09: So, so, yes. So, there are very serious data issues that arise when actually trying to do research. So, access issues, right, for teaching I don't need the particularly specialized data for teaching because what I'm trying to teach the students are some very basic principles about

some very broad topics. But when teaching PhD students, that's basically research. And yes, we do run into problems. So, for example, I have a student who's trying to do work on [topic redacted] and Indian government websites are absolutely horribly run. The Indian government collects a massive amount of census information. And then, doesn't update it, puts it out on websites where it's incomplete. They never have working email addresses if you want to contact them to try and access the micro files, that kind of stuff is a real problem. And there are some universities that have access to these data. And so currently we're trying to get some through [institution redacted]. But I do know that you know if you're at Harvard and places like this, you know, and you want the census files you can get them.

I: Alright, so some of my next questions are going to be related to working with data. So, I would like to understand how do your students manipulate, analyze, or even do the interpretation of the data for your classes, for this particular class. So, what tools or software do your students use? You mentioned Excel, but are there other tools that are used for the statistical analysis or visualization, or any other tools manipulating or working with the data?

UCSB09: No, they use only Excel. You know my philosophy of teaching students to work with data is that, you know, the actual data manipulation is the easy part. And you know that you can sort of pick up in all kinds of ways with online help things and whatnot. What's really difficult to do, is to interpret what you get after you've manipulated it. So I teach them to manipulate data very minimally in very simple ways and think through what they can learn from it. And all that you can just do in Excel, by using tables and simple functions like minimum, maximum, drawing scatter plots, putting lines of best fit through them. And then you know just thinking through that. Of course, you know if my students were doing anything resembling real data analysis, then, then I would use other things I would use STATA, for example, I have run into a problem with masters students and graduate students where they get they sometimes, I've had maybe two or three of them in the last 13 years who got to the point where they want to analyze the data set themselves, and they don't have access to STATA. And so, you know, I've had to buy licenses and things like that for them. So that's one thing that the university could really think about is, is licenses for software that students use, I know that increasingly people are switching to R which is, you know, free. I just haven't made the investment in learning to use R so maybe that's one way I can get around this. But certainly, you know one thing I will emphasize, I've had people say you know why do we worry with Microsoft Excel etc. We should just have everybody using Google. And I don't think that is a good idea, both for these privacy reasons but also because I've never found Google Sheets to be completely satisfactory.

I: Yeah.

UCSB09: Maybe I need to learn, learn more about it but I find the operations for analysis a little bit rudimentary.

I: Yeah. So do you expect your students to have any prior knowledge of tools or software for these classes in particular that you teach for undergrads. Like, do you anticipate them to be at least familiar with Excel prior to starting to class?

UCSB09: No, no I can't. They just don't have that familiarity. In my department we have people who approach the world through literature, through languages so there just isn't that experience. The only thing I will not teach them is arithmetic. So, you know, I will say to them at the start of the class, you know, you have to be able to find the areas of rectangles and triangles and tell me about the equation for a straight line. And do you know your basic fractions and percentages, and actually get a lot of students who say that's not fair. Why should I have to do this, etc. You know that's what we're working with. So, the problem, you know we don't really get into problems with, you know, software anything, I gotta teach them from the beginning.

I: So previously you mentioned that. It's not that challenging to get students manipulating the data but more like to make sense of the data, after and to do some analysis. Could you describe any challenges you have experienced relating to students' ability to work with data for this class?

UCSB09: Well, yeah so, I wouldn't say that it's very easy to get them to manipulate the data. I'm just saying relative to... You see what I was reacting to, you know students will often come to me and say, "I want to learn to use STATA. I want to learn to use R". And my answer, my response to them is, "What do you want to do with it?". That's like coming to me and saying "I want to learn to use an X-ray machine. But, why do you want to learn? What is the question you want to answer? What will you do with these tools?". And, you know, if they start out with oh I just want to learn to program this, that or the other, they will never produce anything of value with it. If they start out with, Here's a question, you know, here are some comparisons I want to make here are some sort of logical conclusions I want to be able to draw from a data analysis, we can design the entire analysis on a blackboard, without ever actually touching any software. And then once they have that, then we can throw it on the software and, you know, make the analysis happen. But otherwise, they're dealing with two challenges at the same time, the challenge of, what am I actually trying to accomplish here with the software and then how do I use the software. And that, so that's where the problem is. So, what I really try to focus on first is showing them how to interpret data. So, the first four weeks of term we're looking at things on the board, where I have analyzed the data with the software, I show them the charts and we talk about what can you learn from this, what are the alternative interpretations etc. Then week 5, 6, 7 I'll get to your question now, in week 5, 6, 7 we go down to the computer lab, and we'll start actually manipulating the data in Excel. And the way to do that, I find because I have untrained TAs often, and I have totally untrained students is that I have to write step by step manuals on, you know, first you do this, then you can right click on that, then you highlight this column. Right. Then you copy and paste, and you drag and drop it, we have to show them everything from the, from the start to the finish.

I: Describing the entire rationale behind it, you know, the statistical analysis first and then moving to work with the tool and the software, right?

UCSB09: Exactly. Exactly. Exactly. And to be honest, I have not tried anything other than spoon feeding them on how to use the software right. I know that there's things on lynda.com etc.

where I could say you go learn Excel and I want you to produce this. And that could work, but it could also backfire and I'm often dealing with students who are quite upset about having to touch numbers and having to work with numbers in the first place, that if I tried something that pedagogically they're not capable of doing then I would really face a backlash. So I can't ask them to learn these tools on their own. I haven't yet had the guts to do that. Maybe I should. So okay, as far as difficulties with software I'll tell you what one of the big difficulties has been in the past. It's that the computer labs are often difficult to access; there are not enough computer labs and during section time when I try to get a reservation sometimes I've not been able to get one. So when I've not had a reservation we have actually tried and said to students, "Look come to office hours, bring your laptops, etc. You know, and we'll work it out". Then you run into the problem that with, particularly those who use Macintosh is they're all using slightly different versions of Excel that have different bugs. And, you know, trying to get a TA who's not comfortable working with data to assist students who are working with five different versions of the software is a real nightmare. So, the truly important thing I would say for, you know, if the campus wants to help us teach students to work with data, we need more of these labs, and we need a reservation system that is a little bit friendlier that we can sort of access far ahead of time, book far ahead of time so that those of us who are actually running classes get first dibs. You know what happens is you'll often find that the labs have been booked up for open hours, but you know open hours should be scheduled after sections.

I: Right.

UCSB09: You can have open hours at any time, you can't have sections at any time.

I: It's a good point. And indeed, whenever we teach people with different systems like PCs and Macs, it's a nightmare. So, I can relate to that. Yeah, different versions of the same software as well. So, it makes it a little confusing. Yeah. So, anything else you'd like to add in terms of difficulties before we move to the next question?

UCSB09: So yeah. I think that's something that, you know, we really should address as a campus in some ways, which is this, this two cultures problem.

I: Right.

UCSB09: So, I don't know if you're familiar with C.P. Snow and his two cultures essay, it's from the 1960s, but it's basically this idea that you know you've got the, the quantitative technical social scientists who imagine themselves a scientist and you've got the humanists. And we have a pretty difficult time communicating with each other. There is what I've said sometimes it's actually open hostility across this divide. And there is, in the old days that wasn't a problem because students could double major, they could go to classes outside their major etc. And so, students were not as badly affected by this as much as faculty and PhD students but undergrads could get a well-rounded education at UCSB, play on both sides of the cultural divide and they'd be fine. Today, that's no longer true because all these majors are impacted and students can't get into classes outside their major. And as a result, we're churning out

students who, if they come from, sort of the more cultural studies side of things or from the humanities, are never exposed to quantitative social science at all. Even if they're coming out of the qualitative social sciences, they get no exposure at all to quantitative social science. And you know on the flipside too, right. You know [majors redacted] don't know anything about literature, etc. And, you know, that actually has a lot to do with how you should interpret quantitative data too, you need a strong background and philosophy before you should ever touch a quantitative data set. And so, I think that's, that's a major problem. And if the university is going to get serious about liberal arts education which means bridging the two cultures, there actually need to be, you know, core classes for freshmen where they're actually required to do a bit of everything. And actually, address this stuff head on, what I have seen frankly, you know, I know they've got these quantitative requirements to check boxes. But I have seen, I've encountered so many students who have checked the quantitative boxes and can't tell me what 4% of 200 is, much less, you know, tell me anything intelligent about, you know how to interpret a scatter plot, and how not to interpret a scatter plot, you know. So, so I think, I think there is this pedagogical work to be done, that would actually require faculty interest, faculty to come together to think about, sort of, you know, theory of knowledge classes, which include exposure the quantitative social science, hands on work early on, and that in turn would mean investments in computer labs. But these are the real problems we need to fix as a campus, I don't think that there's problems with accessing this data set or that data set, that's not the issue. The real constraint is contact with faculty with the time and the incentives to actually set up you know this kind of education.

I: Okay. Oh, thanks for that. So, moving to the next question, I'd like to understand a little bit about the process of, like, teaching students with quantitative data and how it connects to the learning goals in your field. Can you describe how the ways in which you teach with data relate to the learning goals in your discipline at large, or even [discipline redacted], considering that's your background?

UCSB09: Sure. So, I'll talk about [discipline redacted], because frankly my department has more or less rejected quantitative social science. There just isn't any offered there if you read the journals, it's not there. If you look at the faculty, not just at UCSB but in general, anywhere in anywhere in the United States or, outside of Eastern Europe, all the [department redacted] faculties that I've looked at I've seen are entirely non-quantitative. So, I'll talk about that sort of teaching. I would say [redacted topic]. Let's call the field [redacted field] sort of halfway between [redacted fields], right. So, it's interdisciplinary. And it is both quantitative and qualitative and that's what I think I'm teaching my students. So, how does quantitative work relate to the teaching goals. So, there are several things I want students to understand. One of the first things you have to do in teaching any kind of social science is you have to convince students that they're wrong about everything and they're probably wrong about everything. So, they have to be confronted with a variety of kinds of evidence. And they have to be confronted with evidence that refutes their preconceived notions and quantitative evidence can be very good for doing this. Right, because it really forces them to sort of sit down and say, Okay, if this is my view of the world. First of all, you know, in how many places this is relevant? You know, forget about causation, do we even see correlations to suggest that my worldview is in fact the correct

one, or that, or that these particular social forces operate the way that I see they do etc. And so, the one of the nicest things about it is it, you know using quantitative data in classes forces students to stand back and think about evidence in a way that does not presuppose the answer. I don't want to, you know, sort of suggest that quantitative evidence is better than qualitative evidence, that that isn't the argument. But systematic approaches to research are more likely to dispel confirmation biases than non-systematic ones. And the nice thing about quantitative data is that, you know, the only way to analyze it is systematically, I mean, there are people who will try to manipulate it etc., but if you give up on those dirty habits, then it's, you know, it's a systematic approach to data. So, you agree first on how I am going to analyze the data, how will I interpret what comes out, and only then do you get to see what comes out. And the effect of that is that often what comes out to something that makes you think about the world differently. And that's one of the main goals in teaching this kind of stuff but particularly at the undergraduate level.

I: My next question relates to how you do usually approach, like, what's your approach to make your students to think critically about the data sources they might encounter like in their everyday life? Can you expand a little bit about the approach that you usually follow to teach them when they are confronted with evidence and, you know, to assess and interpret the data?

UCSB09: Yeah. So, so there's, you know there's two things here there's the data, and there's the results of the data analysis. So, when I say the data, I mean, you know, the starting numbers that go into the analysis where do those come from. So, in, in my classes, whenever I introduce a variable, first of all, they have to understand, they're obliged to understand every aspect of the definition of the units in which it's measured and the real world process by which somebody arrived at that number. So, for example, you know, if we're going to look at a GDP number right. The question is, what units is it measured in, what does it capture, who made the number, who invented the number, what are the components of the calculation that were imputed, who imputed them. And to bear in mind, and then I always tell them a few stories about people who, who may have made up numbers or used dubious assumptions for some reason having to do with their political objectives or prior beliefs or whatever else.

I: Right.

UCSB09: So, start out by making sure they understand that all numbers are constructed and emphasize to them that that doesn't mean numbers are useless which is unfortunately what they get from some people, but rather that one must be aware of how they were constructed. Take a sensible decision about, you know, whether you accept the quality of these data or not. And then think about, then we get to the analysis, we think about how you could interpret, you know some relationship seen in the data if you believe the data absolutely fine. And then we also say we're looking at what if they weren't. What kinds of bias measurement is in the data that would actually lead you to incorrect conclusions when you interpret the analysis from that data. We also take a lot of time to cover examples of misinterpretations.

I: Right.

UCSB: So, we go through lots of examples of reverse causality, lots of examples of omitted variables, biases, lots of examples of just spurious correlations. You know, I find that humorous examples help a lot. So, you know, things that people did something so thunderously stupid it really sticks in the mind. So that's the approach.

I: Okay, thank you. That was very comprehensive. In terms of going back to institutional support, because you mentioned the issue of not having labs accessible or being able to schedule for lab timing. In the past year, have you observed any policies or cultural changes at UCSB that would influence the ways that you're teaching with data?

UCSB09: There was one set of changes with the scheduling of the computer labs. So, I don't know exactly how this happened, and this is a story from about two years ago because I was on sabbatical last year. And then this year I haven't been, this year I've taught one online class that didn't involve the labs at all. I don't know what I'm going to do for data teaching in the spring. But, I'm forgetting the names of the scheduling software but we used to have scheduling software, where we would put in a request, the request would immediately get elevated to somebody, I think at collaborate, and then, you know, they would sort it out and they write back and say okay you have labs at these times, etc. We went from there to scheduling the computer labs using the same system as being used to schedule the classrooms. It was difficult to get somebody on the other side. And I think also because there has been increasing demand for quantitative subjects and classes among students. We were running into constraints that the labs simply weren't available, and then there was nobody to talk to about it. And you know the response to it was largely well you know go figure it out. You know, what compromised positions can you come to, are there ways you can teach the stuff in office hours, etc. and that was a real problem. I've also, you know, I do sit on a campus technology committee that oversees Collaborate, etc. And I did hear some fairly disturbing things where people kept saying, maybe we don't need the labs everyone can just use Google stuff. And, you know, they can sort this out and they can learn online and they've all got laptops so we don't need to support all this. And I think what that is really missing is the centrality of a teaching space of physical teaching space where students can come in and learn to just kind of jump through the hoops, monkey with the software and get the basic tasks done. You know, for, for, it may not be. Yeah, I mean I just, I just think this is going to be one of these things that if the campus does go in that direction of not supporting centralized learning computer labs, we're not going to be able to teach quantitative pedagogy, to students, particularly those who are quantitatively challenged. And one other related issue that comes up with this you know one of the things that I've actually been doing research on is sort of racial and economic stratification across majors at the UC's including UCSB. And so, we now, we're increasingly seeing since, over the past 15 years, a segmentation where the stem STEM majors are increasingly getting filled up with students who went to better high schools, they are you know less likely to be students of color less likely to be from working class backgrounds. And so this argument that you can, you know, just have students work on this stuff on their own, and we don't need the centralized learning spaces for them is going to make that kind of thing worse.

I: Right.

UCSB09: There's been, ever since the financial crisis, there was a migration of students from the qualitative social sciences and humanities into the quantitative Social Sciences and the STEM fields. And that has put tremendous pressure, I think, on the lab spaces. And, you know, the STEM departments are using these pre majors to decide who gets to take the majors and they are increasingly squeezing out students of color and lower income students. So there needs to be proper infrastructure to support this, to support students, and you know I've actually heard departments, saying oh we need our own dedicated lab so we can, these are departments that are pre majors who say, we need a dedicated lab computer lab for ourselves, so that we can sort of coach students to help them over these barriers. And I think, you know, whether that's a genuine attempt to help or not, I don't know that's a different question. But the fact that they're asking for this tells you that instructional spaces where people can get that, where students can be trained to get their hands dirty with data, or in very short supply on campus. So that's the constraint. I think that we need, we need to hit, we need to hit it hard.

I: Right, thank you. In terms of ethical challenges, do you think that instructors in your field have any ethical challenges in teaching with data? So let's say, you know, sometimes the nature of the data that you're working with, or the topic? Have you ever experienced any challenges, ethical challenges in teaching?

UCSB09: No, because you know, when I teach a data I'm usually teaching about, you know, we're making comparisons between countries, and we're thinking critically about the data, and there's no, so at the country level there's no questions of privacy invasions or anything like that, these are national statistics. You know, yeah, I mean I think if there's an ethical problem about teaching with data, it's that we don't do it enough. And the departments that teach our working class students do it less than others And, you know, there's a big premium on this, right, you graduate from a four year college, if you have learned to work with quantitative data, it actually improves your future employment prospects. And we are systematically depriving, you know, the students that need upward mobility the most, access to that, that's our ethical problem. So we have to think systematically about, about the rationing of students into majors and think about what, and also about the representation of quantitative social science, particularly the interdisciplines on campus. Because unless we get that sorted we're depriving this, that's where these students are and we're depriving them of methodological training of this type.

I: Moving on to the training and support section of our interview. Earlier, you mentioned that there are a few PhD students that sometimes would work with you or assist your students in the class. Do you get any other support, let's say, other instructors, teaching assistants or drop-in sessions that involve other people to help with your students' questions? Because I remember you mentioned that being a solo task, but if you have some help, could you please describe a little bit more about the help you get?

UCSB09: So honestly. You know I don't I don't actually... It's sort of like this right, if you have a student who has done some undergraduate [major redacted] and has, you know, has operated some software and run some regressions, they now need to learn to do this better. This isn't something you can learn through course instruction, you actually have to learn by dropping into

a lab and so on. You know, there's one exception I have a PhD student who has been getting a lot of very nice help from somebody who's in your GIS lab in the library and that has been incredibly helpful, that kind of drop in thing. Because you know neither my student or I use GIS so for those startup costs, that's really helpful. So again, there's sort of two issues, there's training students in how to design research with data. And there's how to actually operate the software. The vast majority of the training time goes into how do you teach them to think about asking questions of data right that research design part? My difficulty is that, you know, to teach quantitative research design, you actually need like two years in the classroom and in seminars watching it done. But because no one else in my department will do it, you know, my graduate students don't have time in their schedules to get that kind of training. So that kind of training is what I'm having to do hands on my own. And so that's just really hard, I mean that's the bit that, like, I don't, I don't see how the campus can fill in. You know, that's just a result of these two cultures problems that, that, you know, if you're in a department for your colleagues don't believe in quantitative social science or don't believe in the value of, you know, hiring a couple of other people to help teach it, and build it into the curriculum. You either do it on your own or you don't advise PhD students. That's, those are the choices.

I: And what about like external sources or learning tools or resources, to your knowledge, are there any ways in which your students are learning to work with data outside the formal coursework. I know it's only one class, and one new, one new course, but have you heard of other resources that they are trying to explore while they go through your course?

UCSB09: So I do run into undergrads who are actually trying these sorts of things.

I: Right.

UCSB09: The best resource is actually a double major in a quantitative social science or in a STEM field. Because, look really this is the sort of like learning to work with data, it's something that you have. It's a craft. And like any craft you have to learn through apprenticeship, and you have to learn through sort of graduated sequences of classes. There is no substitute for that. You can have as many little drop-in help centers as you want. But, you know, and that but that will help you to pick up the pieces, later on. So for example, you know now that I know how to do, you know I know how to do a whole lot of different kinds of quantitative social science, but I don't know GIS, I could drop into GIS and I know what to ask. And we can gradually figure things out. Same thing with graduate students, if they've had, if they've been through a sequence, if there's no sequence you can't solve it. Undergraduates, you know, there's really good courses in the statistics department where you actually get to, you know, turn the data and use it to ask questions so you're doing both design and programming in R etc. So those students that can access that are getting tremendous benefit from it. But those classes are severely overbooked. I have had some students say oh I've gone on lynda.com and learned to use Excel and it was kind of useful. But again, the trick is, you've got to have a project that has a set of tasks that you have to complete. Right, so that you're actually learning to apply the software to solve an actual problem. You can't learn to use statistical software to solve problems without solving problems. And therefore, you can't learn to use it at all because what's the point

of using it if you're not solving a problem so there has to be, this is why I'm saying pedagogically you want to fix this, you got to have courses that students take, sequences of courses that they take in which they encounter problems and then use software to solve those problems. The other stuff, it exists, they use it some, but it's not real education. Right, it's plugging in a few weeks here and there but it's not getting the job done. YouTube is a great source if you've got, if you've got no problem with, you know, I need to get Excel to do this then you can find a video that will be a YouTube video that will help you to do that. But that isn't quantitative analysis, that is just making some manipulation in a piece of software. And it's not what's going to get our students, you know, elevated into, into positions where you know they can make a living doing this, or where it is even part of their profile. You know on the job. So, I do think we've got to get serious about it in a systematic way. Sorry, I know you're looking for particular kinds of answers and I'm sitting over here answering my own question, and it's irritating when people who are interviewed do it.

I: I think it's great because you're actually suggesting a few things and, yeah, so it's going even beyond, so it's really great because we really want to hear and know not only the current stage or things that instructors are struggling with, but also to understand and like, you know, see some possible solutions. So, it's really nice to hear your insights about these topics. The next and final section of the interview, and I want to check if we can go a little bit over time, if that's fine with you...

UCSB09: It's fine.

I: So, this question is about your own training. So, have you received training in teaching with data other than your graduate degree, for example workshops, technical support, or even like close help from peers? Can you explain a little bit about your own training with statistical analysis, and how you went about learning and advancing your knowledge in this field?

UCSB09: So, so I learned to do statistical analysis in graduate school. And I learned, you know, so I learned a lot by, so I went through the sequences, went through a lot of seminars etc. that was always very useful. The two greatest learning experiences, one was writing, working on working on a couple of papers and writing them with my advisor. Because again it's about sort of seeing how you design an analysis to solve a problem, right, to answer a question. So, that was extremely useful and then of course writing my own dissertation where I then had to apply this in my own work, and come up with my own analyses, etc. That was extremely useful. And then you know you, then you, you know once you're beyond your PhD then you start to practice, and you get feedback on your work and those feedback mechanisms are extremely important. At UCSB, my main support mechanism are the [department redacted] here. There's some really lovely faculty members who are very sharp analysts, very very good at causal inference. So, I go over and get a lot of help there. Well, I wouldn't exactly call it help as much, so on a paper if I hit a roadblock, you know where I know what the right thing to do is, I just can't figure out if it's good enough or how to frame something. I often get help with that. I take my work and present it there and get critique and that of course is extremely useful, because the, you know what, what people in my department think of as good work and what they think of as good work is very

different. Right. Again, it's a two cultures question. So that's, that's how I learned to do statistical analysis, so from doing that, from presenting. Yeah, so it's all learning by doing and you know feedback from people who are, you know, either peers, or people who are even senior so earlier on when I was an assistant professor. You know I was getting schooled by them but you know now I just present work and we talk as peers I get feedback, so that's been good. The learning to teach with data, you know that that's something I've entirely have to figure out on my own. And, you know, I get a lot of practice with that, because nobody else in my department teaches quantitative anything. So, I've had to do a lot of it, and I just sort of learned through trial and error there are, you know, what I recommend for that is actually books written by people who are good at it, not about how to do it but actually demonstrating it. So, you know, there are a lot of books written by, you know, good economists, sociologists, who can take technical material out of journals and express it in a way that is clear and easy to follow. And just reading a lot of those books, helps you to understand what it takes to make what seems like technical material accessible to lay people.

I: Okay.

UCSB09: And so that that's actually the best, those are some of the best examples for me.

I: In terms of exchanging assignment plans or syllabi or other instructional resources, have you ever done that with those departments you collaborate more closely with?

UCSB09: No, I can't. I can't because the people I collaborate closely with are in [discipline redacted], and they can expect their students to know some things. So, their assignments simply wouldn't work for my students. And as I said in my department there's nobody who would actually do anything quantitative in class.

I: And what about outside UCSB, with other universities?

UCSB09: Honestly I've never really needed to do that. I mean I've done this enough that I've sort of got a knack for it. You know there are, every once in a while, I run across something on YouTube where there's somebody who's actually you know, managed to teach people about data and a very creative way. So, again, you know I mentioned gap minder. You know, the founder of gap minder, he's dead now but he was this Norwegian public health guy who does extremely, extremely useful videos and you know shows you how you can use animated data to explain very simple things to students, I've been trying to push it a bit harder to use animated scatter plot to explain more complicated things. And it works but I mainly do this through trial and error. I don't get it from other people. Again, because there are very few places, and very, you know, because we live in this bifurcated world where either you do quantitative work or you to qualitative work, there are actually very few places where you have someone trying to teach students to do quantitative work where those students will never before, and never again see quantitative work. Usually, you can assume that there is a sequence of there's something before or there'll be something after. And because that never happens here, I have to make it all up.

I: Okay. Considering evolving trends in your field, what types of training, or assistance would most be beneficial to instructors in teaching with data? What would you like to see, you know, being offered on campus for you to teach with data? You mentioned a few things but if you can explore it a little bit more that will be helpful.

UCSB09: When you say offered, are you talking about courses, are you talking about labs, or are you talking about infrastructure or support structures. What do you mean?

I: Like the types of training or assistance, it can be broadly understood as infrastructure, or even people to help. You know, what are types of training, or things that could be offered?

UCSB09: Okay, so I've heard, I've heard periodically so there's. Okay, so, you know, my position is you don't teach people to work with data, unless you teach them how to use data to solve problems or to answer questions right, so you have to do that. That takes far more than just sort of dropping into a lab or a few class sessions that takes a sequence. So that's what needs to be there. Now there's two related things here, one, one issue that we have at UCSB is that if somebody wants to learn to program a computer, and they're not a computer science major there is no class they can take. So, all the programming classes are restricted entry. This has to change; this is a very serious problem. And the related, so, and I'll tell you where I'm going with this. You know, if we need to have sequences where people learn to solve problems and answer questions using quantitative data. And if we need to expand our access to training in, not just social scientific data analysis of the sort of classical variety that I'm doing but also the kinds of Bayesian analysis, machine learning and things like that that students increasingly want to have because that's the current niche in the labor market. The way we solve this is by actually having, you know, either data sciences or a quantitative social science minor of some sort. Right, where we actually provide FTE (referring to full-time equivalent) for it. Right, so the campus, the campus could reach out and find faculty in other departments, right, including people like me. Right, who would absolutely love to spend you know one quarter of the four that we teach, right, one of our teaching courses would be actually teaching students to apply data to solve social science problems. So, if we could do that, it would be incredibly popular, the demand is absolutely there, we would have to, you know, work at the campus level with the various deans, who would have to find the faculty FTE somewhere to make this happen. We would have to back it up with computer labs. But I think if we got a bunch of us like-minded people around the table and we're actually told there will be the resources to make this happen, we could do great things. But that's, it's going to take sequencing, you can't ad hoc this, this is a pedagogical issue and it has to be handled as a pedagogical issue, it has to be handled with sequences, problems, and resources.

I: Thank you. Yeah, wrapping up. So is there anything else from your experiences or perspectives as an instructor on the topic of teaching with data more broadly that you would like to share with our group.

UCSB09: Yes, I think in addition to this, so that I've talked about two different pedagogical models here the one I was just talking about is you can have a minor, where students who want

to get serious with this stuff self-select and then they kind of proceed and they go through it. I think that's really useful, it's something to have. The other thing to have, is for freshmen to have, like I said, some kind of a core, a core knowledge sequence that students coming in so not, you know, change the way that we do, GE (referring to General Education requirements), I think, the way that we do GE is a bit scattershot. The general education credits. If you actually have, you know, some sequences the freshman year, taught by, you know, really well regarded instructors, find the ones who've been getting the teaching awards etc. across campus, put them in the classroom for freshmen, make them work across the two cultures. So, you have some classes that are a little quantitative, some that are a little bit qualitative, some that are both and ensure that students coming in, who say things like I'm not a numbers person, etc. are forced to actually reconsider that position. So that early on, they can make decisions and actually start to get into the kinds of majors that they're actually well suited to. And you know that actually interests them. I'll tell you. Almost every time I teach my [class code redacted], I have a few students in the class who come to me and say I'm struggling with this, I'm not a numbers person, I shouldn't have to do this and please excuse me I won't learn this material, and almost every single one of the people who says this is female. Because that's what we do as a culture right, we've historically toward women they can't do math. And it's rubbish, all the empirical evidence says that's rubbish. And you know the other group of students I have who come into my classes are students of colors, working class students who say I wanted to major in [major redacted], and I couldn't do it because I couldn't hack the math etc. But I know that it'd be actually really good at it. If they actually had the support systems early on. So, starting from freshman year. We've got to start building the support systems to teach applied mathematics to students of color, and to women in particular and I'm not saying target them but like make it something everyone's got to do so that everyone starts from the same basis. And maybe we'll start to get some of this stuff sorted. So, a sort of general education core knowledge approach that incorporates quantitative material in a challenging way, you know students shouldn't feel like they have to study hard or they might fail it and it'll be a problem.

I: I see. Well, thank you so much. Anything else. Sorry if I cut you off.

UCSB09: No, not at all. I just feel like we are depriving a lot of our students from things that their more privileged peers are getting, and then we try to put a band-aid on it and it's not working.

I: Thank you.